CO1_ASSESSMENT-01

CONCEPT MAP

ITA0329 – MOBILE COMPUTING

Grace Johnson M
192472044

14.Develop a concept map that evaluates the complete mobile communication process.Illustrate the interaction among user equipment, access network, core network, mobility management,and service delivery.

Determine the most critical stages in ensuring communication quality.

Justify your evaluation with meaningful linking phrases.

Complete Mobile Communication Process

Main Components of Mobile Communication

Five Core Components

User Equipment (UE)
- Initiates communication
- Converts voice/video to data
- Transmits/receives radio signals
- Uses SIM/eSIM
- Implements communication protocols
- Measures radio conditions

Access Network
- Establishes radio connectivity
- Allocates radio resources
- Transmits/receives wireless info
- Manages radio scheduling
- Measures signal strength/quality
- Supports handover
- Manages congestion/failures

Core Network
- Authenticates subscribers
- Establishes/manages sessions
- Routes user data
- Connects to internet/other networks
- Applies subscriber policies/QoS
- Supports mobility/session continuity
- Collects charging info

Mobility Management
- Registration
- Location management
- Paging
- Handover
- Reachability management
- Session continuity

Service Delivery
- Voice/video calls
- Web browsing/internet access
- Messaging/multimedia sharing
- Video streaming/online gaming
- Cloud apps/business services
- IoT applications

Communication Process Stages

Seven Key Stages

- Stage 1: Communication Initiation
- Stage 2: Radio Access
- Stage 3: Authentication & Security
- Stage 4: Core Network Processing
- Stage 5: Mobility Management
- Stage 6: Service Delivery
- Stage 7: Response Transmission

Interactions & Dependencies

- User Equipment <-> Access Network
- Access Network <-> Core Network
- Core Network <-> Service Delivery
- Mobility Management <-> Access Network
- Mobility Management <-> Core Network
- All Components & QoS

Most Critical Stages for Quality

Ranked Criticality

- Rank 1: Access Network & Radio Link
- Rank 2: Core Network & Session Management
- Rank 3: Mobility Management & Handover
- Rank 4: Security & Authentication
- Rank 5: Service Delivery & App Performance
- Rank 6: User Equipment

Real-World Examples

- Weak Network Signal
- Moving Between Cellular Towers
- Strong Signal but No Internet
- Incoming Call to an Idle Device

Strategies for Improving Quality

- Improve Access Network coverage
- Reduce interference/congestion
- Improve Core Network reliability
- Optimize handover procedures
- Strengthen security
- Improve service delivery
- Apply Quality of Service management

Conclusion

- The Access Network
- Mobility Management
- Service Delivery
- User Experiment